RAJALAKSHMI ENGINEERING COLLEGE

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A2

Started on	Wednesday, 19 August 2026, 8:50 AM
State	Finished
Completed on	Friday, 4 September 2026, 7:05 PM
Time taken	16 days 10 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Recursive function to find two elements whose sum equals x
int findPairRecursive(int arr[], int left, int right, int x, int *a, int *b) {
    // Base case: no pair found
    if (left >= right) {
        return 0;
    }

    int sum = arr[left] + arr[right];

    if (sum == x) {
        *a = arr[left];
        *b = arr[right];
        return 1; // Pair found
    }
    else if (sum < x) {
        // Move left pointer to the right
        return findPairRecursive(arr, left + 1, right, x, a, b);
    }
    else {
        // Move right pointer to the left
        return findPairRecursive(arr, left, right - 1, x, a, b);
    }
}

int main() {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) {
        return 0; // Invalid input
    }

    int *arr = malloc(n * sizeof(int));
    if (!arr) return 0; // Memory allocation failed

    for (int i = 0; i < n; i++) {
        if (scanf("%d", &arr[i]) != 1) {
            free(arr);
            return 0; // Invalid input
        }
    }

    int x;
    if (scanf("%d", &x) != 1) {
        free(arr);
        return 0;
    }

    int a, b;
    if (findPairRecursive(arr, 0, n - 1, x, &a, &b)) {
        printf("%d\n%d\n", a, b);
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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